Midterm (practice version - NOT GRADED), Math 170S, Spring 2019
Instructor: Elizaveta Rebrova

Printed name: ____________________

Signed name: ____________________

Student ID number: ____________________

Instructions:

- Read problems very carefully. If you have any questions please ask.
- You should show all the details of your solution, the correct final answer alone gets a very small part of the credit. If you do not have time to write a proper solution, always attach your drafts and try to make them easier for us to follow.
- You final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Calculators and books are not allowed. Phones, computers and any other communication devices are not allowed too.
- You can bring a letter size piece of paper with formulas and anything helpful for you (please be reasonable and do not squeeze out the whole textbook inside one page :)). Please bring your own writing utensils. We provide all necessary distribution tables and extra draft paper. If you need more draft paper please raise your hand.
- Please have your ID ready so we can check it without disturbing your work.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. *For this problem: provide short explanations only.*

   In a certain political campaign, more then 50% supportive votes will secure a win of a candidate. One candidate has a poll taken at random among the voting population, and computed confidence intervals (for the fraction of voters in favor) with confidence level $1-\alpha$: a one-sided interval $[a, b]$ for the lower estimate; and a two-sided interval $[c, d]$.

   (a) (2 points) What is b?

   (b) (2 points) What can be bigger, a or c?

   (c) (2 points) What condition must be satisfied so that a candidate feels confident about her win? (with probability $1-\alpha$)

   (d) (2 points) (Not about these elections any more :)) $X_1, \ldots, X_n$ is a sample from the distribution $Unif[0, \theta]$. Is the estimator $\tilde{\theta} = 2\overline{X}$ biased or not?

   (e) (2 points) True or false: in the example above, if $X_{(n)}$ is a sufficient statistic, then $X_{(n)}^2$ is also sufficient?

   Hint: what are the possible values of θ?

2. (10 points) You have a biased coin with Θ being a probability of heads. You believe that Θ has pdf

$$f_\Theta(\theta) = 2 - 4|0.5 - \theta| \text{ for } 0 \leq \theta \leq 1$$

Find the MAP estimate of Θ, if you observed exactly one heads in n experiments? (Get a general formula for any $n = 1, 2, \ldots$)

Check your answer: what is the best $\hat{\theta}$ you got

- for 5 experiments (coin flips)?
- for 2 experiments (coin flips)?

3. Consider a sequence of independent coin tosses, and let θ be the probability of heads at each toss.

   (a) (5 points) Let k be a fixed number, and N is the number of tosses until k-th head occurs. Find maximum likelihood estimator of θ (depending of N).

   (b) (5 points) Now let n be a fixed number, and K is the number of heads in n tosses. Find maximum likelihood estimator of θ (depending on K).

4. (10 points) Let $X_1, X_2, \ldots, X_n$ are samples taken from the uniform distribution

$$Unif([\theta, \theta + 1]).$$

What is the length of a confidence interval with confidence coefficient $1 - \alpha$ for the estimate $X_{(1)}$ for parameter θ?

5. (10 points) Let $X_1, X_2, \ldots, X_n$ be samples taken from some distribution with unknown density function. However, you know that $\mathbb{E}X = \alpha/\beta$ and $\operatorname{Var} X = \alpha/\beta^2$, where α and β are two parameters of the density function (and there is no third parameter). What method could we use to find good estimates for α and β? What are these estimates?